

Caffevolve — Phone-Primary Personalization

Addendum to “The Intelligence Center as a Distributed Process”: where profiles, plans and routines are created and edited, and how edits stay in sync

Prepared for: Brent Langevin **Purpose:** Define which personalization tasks are created on the phone and which on the device, how edits move between them, and the commitments that keep the device useful on its own. **Status:** working draft, revised after review. The rules below were agreed in review; section 12 lists what is still open.

1. The core rule, in one line

The phone is the authoring surface; the device is the execution surface. Heavy authoring starts on the phone, and both surfaces may adjust what exists. All brew, grind and drink-path decisions, and every brew initiation, remain at the device. Personalization can be done wherever the user is comfortable.

2. Why

Personalization has grown: the initial profile, the Caffeine Behavior Instrument (CBI), subprofiles, BrewNow profiles, additive and grind preferences, and now daily plans and routines. Doing all of it at the device means long sessions standing at a countertop. Moving the heavy, optional authoring to the phone removes that friction and should reduce reluctance to use the personalization tools. The architecture already supports it: the engine logic is compiled into the app, and the per-user cloud store already carries the event log and the user’s parameters between the two copies of the engine.

3. What is created where

Item	Create	Edit
Basic Resident Profile (identity and the CBI-derived Caffeine Comfort Zone: Peak-Zone Floor, Wake-Hours Ceiling, Bedtime Ceiling, Caffeine Decay Rate)	Device or phone. Phone preferred. Resumable.	Either, through the profile review screen
Subprofiles; BrewNow profile; additive preferences; grind setting; workplace settings	Phone only	Either
Plans and routines (Plan my Day)	Phone only	Phone (full). Device: skip today’s plan or make light edits; Plan Check runs at the device.

Item	Create	Edit
Guest profile (Quick Profile Path)	Device only	Device only

Each item is created in one place at a time, which removes most conflict risk. Remaining conflicts occur only on edits and are covered in sections 6 and 7.

4. The Basic Resident Profile

The CBI is part of the Basic Resident Profile, because it produces the profile’s core parameters. A resident profile without a completed CBI is not an established profile.

- **Complete on the device.** A resident can unbox the machine and finish a full Basic Resident Profile, CBI included, with no phone and no account. The device is never unusable for want of a phone.
- **Phone preferred.** The CBI (twelve questions) and the rest of setup are the longest single session. The device first-run screen says so and offers to continue on the phone.
- **Resumable.** The setup flow already offers “Exit setup & complete later.” A user may answer part of the CBI on one surface and finish on either. The profile counts as established only when the CBI completes; until then the device uses conservative guest-style defaults.
- **Bedtime is not part of the profile.** It is a session value, set and revisited at the device with each brew. It is therefore not synced and cannot conflict.
- **Phone-only layer.** Subprofiles, BrewNow profiles, additive and grind preferences, and plans require the phone to create. On the device these screens say “available in the CaffevolveGo app,” and do not appear broken.

5. The boundary model

The Caffeine Comfort Zone has four parameters. Users can adjust only the two ceilings.

Parameter	Source	User-adjustable
Wake-Hours Ceiling	CBI, then refined by the AFS	Yes, any time, from the profile review screen (“Fine-tune ceilings”)
Bedtime Ceiling	CBI, then refined by the AFS	Yes, same as above
Peak-Zone Floor	CBI, then refined by the AFS	No
Caffeine Decay Rate	CBI, then refined by the AFS	No

- **Each ceiling has three parts.** The *computed value* (from the CBI, moved by the AFS); an optional *user override* (set with the slider); and the *effective value*, which is the override if one exists, otherwise the computed value. The device brews against the effective value. “Use computed” removes the override.

- **The platform only advises.** Setting a ceiling higher than most people are comfortable with shows an advisory message, not a block. The user decides.
- **The AFS never overwrites an override.** It continues to refine the computed value in the background, so “Use computed” returns the learned value. Its boundary-triggered prompts fire around the effective value.
- **Refinement is stored as feedback events.** Each feedback response is an immutable event with a source and ID, and the computed value is derived from the baseline plus those events. Events arrive from either surface and merge as a simple union, in any order, so the two surfaces cannot disagree about the computed value once they hold the same events and the same engine version.

6. Editing and conflicts

The existing principle carries over: state changes are appended, not edited in place. Applied to profiles, subprofiles, preferences and plans:

- Every edit produces a **new version of the record**, stamped with its source (phone or device), time and the version it was based on.
- If both surfaces edited from the same base version, the **later edit wins per field**. A ceiling change on one side and a rename on the other both survive.
- If both changed the **same field**, the later edit wins and the other surface shows a one-line notice that its earlier change was replaced.
- The **two ceilings are treated as a group**, because they are adjusted together. If both surfaces edited them, the later edit wins for the pair, so the result is never a mix the user did not see.
- Each surface shows **sync state** (for example, “Synced to your Caffevolve 2 min ago”), especially after a ceiling change, because the device brews against whatever it last received.

For brew events the device remains the source of truth. For personalization records, the most recent valid version is authoritative, whichever surface wrote it.

7. Edit Profile: re-running setup

From the profile review screen, **Edit Profile** re-runs the whole setup flow, and everything in it can be changed: the CBI answers, subprofiles, grind setting and additive preferences. It is available on either surface. A re-run has two kinds of outcome:

- **CBI answers unchanged.** Only the other fields change, and they merge as ordinary field edits. The user’s ceiling overrides and the AFS learning are kept. Changing an additive must not wipe tuned ceilings.
- **CBI answers changed.** This is a new baseline. The computed boundaries and decay rate are recomputed, the user’s **ceiling overrides are cleared**, and the **AFS learning starts over**. The user passes through the fine-tune screen again and may set new overrides.

- Any edit made on the other surface against the old baseline is superseded by the new baseline, and that surface shows a notice.

8. First run, pairing and households

- A new device with no resident profile offers two paths: build a Basic Resident Profile now at the device, or continue in the app. The screen shows a pairing code or QR that both pairs the device and signs in the account.
- A profile created on the device stays **local until it is claimed** by an account. Unclaimed profiles work normally at the device.
- A shared device can carry several resident profiles. **Each profile is claimed by exactly one account**, with the person choosing which profile is theirs when they pair. Pairing must never attach every profile to whoever pairs first.
- On claiming, the app **adopts** the device's profile into the account. It does not create a second one. If the person has already created a profile in the app, the app offers to adopt the device profile or keep the account profile, and never silently makes a duplicate.
- A local-only profile has no cloud backup, and a device reset or failure loses it. The app says so when offering adoption.
- Resetting or transferring a device wipes its local profiles and unpairs it.

9. Guests

The phone-first rule applies to Resident Users only. The Guest path stays entirely on the device: name, initials, self-selected sensitivity level and bedtime, with no account and no CBI. A Returning Guest Profile continues to be kept for 72 hours.

10. Safety and access

- **No PIN on profile edits.** The only PIN in the prototype is the Device Administrator PIN, which guards device settings (Wi-Fi, canisters, hardware). Brewing and user profiles do not require it, on either surface.
- The platform continues only to advise. The user decides what to brew and what their ceilings are.

11. Design commitments

- **Minimal on-device path.** The device always supports the Basic Resident Profile and the Guest path, so it works without a phone.
- **Full local copy.** The device holds a complete copy of the active profiles and plans, so a cloud or app outage never strands a user's setup. Brewing continues from the last synced state.

- **The cloud remains a store, not a brain.** It holds the record and the current versions. It does not compute residual or decide what to brew.
- **Data governance defined before launch.** CBI answers describe sleep and caffeine habits and are health-adjacent. Consent, retention, export and account deletion rules need to be set, in addition to per-user isolation and encryption.
- **Version matching extended.** The requirement that firmware and app carry the same engine version also covers the record schema and the shared personalization flows, so older firmware does not meet newer app flows unprepared.
- **Validate the assumption.** Test completion rates for the CBI and planning on a phone prototype against the device before committing the full build.

12. Open items

- Confirm that the Peak-Zone Floor is not user-adjustable (the prototype offers fine-tuning only for the two ceilings).
- The exact set of light plan edits allowed at the device.
- The pairing and account sign-in flow, and what happens on device resale.
- Plan my Day (FIG. 43A, step 3802) lists bedtime as loaded with the profile, but bedtime is a session value. Decide where planning gets its bedtime, particularly when planning on the phone, and adjust the figure.
- Add a phone authoring entry point to FIG. 43A.
- The platform reference mentions a resident profile PIN, while the prototype has none. Reconcile the reference text.
- The prototype does not yet model AFS learning state, sync or pairing; these are specified here but not yet shown.

This addendum records the direction as agreed in review. Items in section 12 are not yet settled.